

Two-Factor CIR Model Estimation

1 Goal of the Kalman Filter in Multi-Factor CIR Estimation

In a **two-factor CIR model**, we have two latent state variables,

$$x_{1,t} \quad \text{and} \quad x_{2,t}. \quad (1)$$

These follow continuous-time stochastic differential equations of the form:

$$dx_j(t) = \alpha_j [\beta_j - x_j(t)] dt + \sigma_j \sqrt{x_j(t)} dW_j(t), \quad j = 1, 2. \quad (2)$$

Meanwhile, we observe **bond yields** at discrete times $t = 1, \dots, T$ for various maturities τ . The yields are linear functions of the states in **risk-neutral** form, with "speed" parameters $\alpha_j + \lambda_j$. We typically denote:

$$Y_t(\tau_i) = -\frac{1}{\tau_i} \sum_{j=1}^2 \ln A_j(\tau_i) + \frac{1}{\tau_i} \sum_{j=1}^2 B_j(\tau_i) x_j(t). \quad (3)$$

A **Kalman Filter** is used here to:

1. **Infer** the latent states $x_{1,t}, x_{2,t}$ at each date t from the observed yields (measurement step).
2. **Accumulate** the probability (likelihood) of these observations under a (quasi)-Gaussian assumption, so we can perform **Maximum Likelihood Estimation (MLE)** of the model parameters $\{\alpha_j, \beta_j, \sigma_j, \lambda_j\}$ **and** the measurement noise.

Since a CIR model is *not perfectly linear* (the variance depends on $\sqrt{x}$), the code below uses an **approximate** or **extended** Kalman Filter method. Nonetheless, it follows the standard KF structure of "predict" and "update" steps.

2 Main Variables and Steps

2.1 State Vector

$$\mathbf{x}_t = \begin{pmatrix} x_{1,t} \\ x_{2,t} \end{pmatrix}. \quad (4)$$

We treat it in **discrete time** with step size Δt .

2.2 State Transition

$$\mathbf{x}_{t+1} = F\mathbf{x}_t + \mathbf{c} + \varepsilon_t, \quad (5)$$

where:

- $F \approx \text{diag}[e^{-\alpha_1 \Delta t}, e^{-\alpha_2 \Delta t}]$.
- $\mathbf{c} = [\beta_1(1 - e^{-\alpha_1 \Delta t}), \beta_2(1 - e^{-\alpha_2 \Delta t})]^\top$.
- ε_t is an approximate **Gaussian** noise with covariance Q_t .

2.3 Measurement Equation

$$\mathbf{Y}_t = A_{\text{vec}} + B_{\text{mat}} \mathbf{x}_t + \eta_t, \quad (6)$$

where:

- $A_{\text{vec}}, B_{\text{mat}}$ are computed from the CIR **closed-form** solution for yields.
- η_t is measurement noise, modeled as $\mathcal{N}(\mathbf{0}, R)$.

2.4 Kalman Filter Steps

- **Predict Step:**

$$\hat{\mathbf{x}}_{t|t-1} = F \hat{\mathbf{x}}_{t-1|t-1} + \mathbf{c}, \quad (7)$$

$$P_{t|t-1} = F P_{t-1|t-1} F^\top + Q_{t-1}. \quad (8)$$

- **Update Step:**

$$\nu_t = \mathbf{Y}_t - (A_{\text{vec}} + B_{\text{mat}} \hat{\mathbf{x}}_{t|t-1}), \quad (9)$$

$$S_t = B_{\text{mat}} P_{t|t-1} B_{\text{mat}}^\top + R, \quad (10)$$

$$K_t = P_{t|t-1} B_{\text{mat}}^\top S_t^{-1}, \quad (11)$$

$$\hat{\mathbf{x}}_{t|t} = \hat{\mathbf{x}}_{t|t-1} + K_t \nu_t, \quad (12)$$

$$P_{t|t} = (I - K_t B_{\text{mat}}) P_{t|t-1} (I - K_t B_{\text{mat}})^\top + K_t R K_t^\top. \quad (13)$$

3 Log-Likelihood

Each time step t , the innovation ν_t follows $\mathcal{N}(0, S_t)$. The log-likelihood accumulates as:

$$\ell_t = -\frac{1}{2} \left[\ln |S_t| + \nu_t^\top S_t^{-1} \nu_t + N \ln(2\pi) \right]. \quad (14)$$